INVESTMENT COMMITTEE MEMO

U.S. REIT Performance and the 15-Year Mortgage Rate Environment

QM 2023 Capstone — Spring 2026 | University of Tulsa

Team: Lailah · Stevie · Isabelle · Sienna

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1. Executive Summary

We analyzed 286 U.S. REITs over 2000–2023 (2,428 firm-year observations) and found that rising 15-year mortgage rates are strongly associated with lower REIT market valuations but do not produce a statistically detectable effect on annual total returns. Specifically, a one percentage-point increase in the 15-year mortgage rate (lagged two years) is associated with a –0.1141 decline in log market equity ($p = 0.002$), implying that the 2022–2023 rate-hiking cycle of ~5.25 pp compressed REIT valuations by roughly 45% in level terms.

Recommendation: Given that rate sensitivity is concentrated in valuations rather than near-term income returns, we recommend a defensive tilt toward REITs with lower leverage and shorter-duration lease structures—specifically Industrial and Net-Lease names—while underweighting highly leveraged Office and Retail REITs until the rate environment stabilizes.

2. Methodology

2.1 Data Sources

Primary dataset (REIT Master): REIT_sample_2000_2024_All_Variables.csv — annual REIT-level observations containing total return (*usdret*), market equity, assets, debt/assets, cash/assets, ROE, book-to-market, and beta. Cleaned via code/fetch_reit_data.py; output: data/processed/reit_clean.csv.

Supplementary dataset (FRED): MORTGAGE15US.csv — weekly U.S. 15-year fixed-rate mortgage series from the Federal Reserve Economic Data (FRED). Annualized to a calendar-year average via code/fetch_fred_data.py; output: data/processed/fred_clean.csv.

Final analysis panel: data/final/reit_fred_analysis_panel.csv — 2,965 rows × 26 columns covering 286 unique REITs over the 2000–2023 period (24 years). Merged via code/merge_final_panel.py.

2.2 Sample Construction

- N_REITs = 286 unique firms; N_years = 2000–2023 (24 calendar years).
- Returns model sample: 2,428 complete cases (missing values excluded listwise).
- Log market equity model sample: 2,429 complete cases.
- Key exclusions: observations with missing core financial controls or missing mortgage-rate lag.

2.3 Model Specifications

Model A — Entity Fixed-Effects Panel (main): Annual REIT return or log market equity regressed on the two-year lagged mortgage rate, firm-level controls (assets, debt/assets, cash/assets, ROE, book-to-market, beta), a linear year trend, and crisis-period indicators (2008–2009, 2020–2021, 2022–2023). Standard errors are clustered by entity *and* year to account for the heteroskedasticity detected via the Breusch-Pagan test (LM $p < 10$■■■). Year fixed effects are replaced by a trend + crisis dummies because the national mortgage-rate regressor is perfectly collinear with full year dummies (verified via strict TWFE absorption check).

The two primary estimating equations are:

Eq. (1) — Returns: $Return_{it} = \alpha_i + \beta■\cdot MORT\_LAG2_t + \beta■\cdot TREND_t + \beta■\cdot CRISIS_t + \gamma\cdot X_{it} + \varepsilon_{it}$

Eq. (2) — Log Equity: $\ln(MKTCAP_{it}) = \alpha_i + \beta■\cdot MORT\_LAG2_t + \beta■\cdot TREND_t + \beta■\cdot CRISIS_t + \gamma\cdot X_{it} + \varepsilon_{it}$

Model B — Random Forest Predictive Benchmark: Year-based train/test split (train: 2002–2018; test: 2019–2023) using the same feature set. Included as a predictive benchmark against linear OLS; neither model outperforms a naive mean baseline (test R^2 both negative), confirming that the FE model is interpretive rather than predictive in nature.

2.4 Variable Definitions

Variable	Definition
MORT_LAG2	U.S. 15-year fixed mortgage rate (FRED: MORTGAGE15US), 2-year lag
usdret	Annual total REIT return (USD, fractional)
ln(MKTCAP)	Natural log of annual average market equity (USD millions)
TREND	Linear year trend (year – 2000)
CRISIS 08-09	Indicator = 1 for fiscal years 2008 and 2009
CRISIS 20-21	Indicator = 1 for fiscal years 2020 and 2021
TIGHTEN 22-23	Indicator = 1 for fiscal years 2022 and 2023
debt/assets	Total debt divided by total assets (leverage ratio)
cash/assets	Cash and equivalents divided by total assets
beta	Market beta (systematic risk measure)
book-to-market	Book value of equity divided by market equity

3. Results

3.1 Table 1: Main Fixed-Effects Regression Results

Both outcomes modeled with entity FE, year trend, crisis controls, and two-way clustered SE.

Variable	Returns Model Coef. (SE)	Returns p-value	Log Mkt Equity Coef. (SE)	Log Mkt Equity p-value
Mortgage rate lag 2	–0.0007 (0.0050)	0.896	–0.1141*** (0.0370)	0.002
Year trend	–0.0006 (0.0015)	0.697	0.0573*** (0.0113)	<0.001

Average assets	–0.0000 (0.0000)	—	0.0001*** (0.0000)	<0.001
Debt / assets	–0.0050 (0.0098)	0.611	–1.4903*** (0.3793)	<0.001
Cash / assets	–0.0291 (0.0184)	0.115	–1.4567*** (0.3695)	<0.001
ROE	–0.0004 (0.0005)	0.420	–0.0053 (0.0134)	0.694
Book-to-market	0.0053 (0.0059)	0.370	–0.4892*** (0.0728)	<0.001
Beta	0.0136*** (0.0049)	0.006	–0.2414*** (0.0645)	<0.001
2008–2009 crisis	–0.0081 (0.0235)	0.731	0.0620 (0.0858)	0.470
2020–2021 crisis	0.0064 (0.0157)	0.683	0.0270 (0.0527)	0.609
2022–2023 tightening	–0.0134 (0.0128)	0.295	–0.2102** (0.0997)	0.036
Entity FE	Yes		Yes	
Time FE	Trend + crisis dummies		Trend + crisis dummies	
Clustered SE	Entity + year		Entity + year	
N	2,428		2,429	
Within R^2	0.075		0.689	

**p<0.10 **p<0.05 ***p<0.01 (two-way clustered standard errors, entity and year).*

3.2 Interpretation of Main Results

The mortgage-rate coefficient in the returns model is –0.0007 with a p-value of 0.896, which is statistically indistinguishable from zero at any conventional significance level. This means we cannot conclude that 15-year mortgage rates reliably predict annual REIT returns after controlling for firm fixed effects and time trends.

By contrast, the log market equity model yields a coefficient of –0.1141 ($p = 0.002$). This implies that a one percentage-point increase in the 15-year mortgage rate (lagged two years) is associated with an 11.4% decline in REIT market capitalization, holding firm fixed effects and time trends constant. Applied to the 2022–2023 rate-hiking cycle of approximately 5.25 percentage points, this translates to an implied valuation compression of roughly 45% in level terms—a magnitude consistent with observed REIT index drawdowns during that period.

Beta is the only control that is significant in the returns model (coef = 0.0136, $p = 0.006$), suggesting higher-beta REITs earn slightly higher average returns. In the equity model, leverage, cash holdings, book-to-market, and beta all load significantly with expected signs, and the high within-R^2 (0.689) confirms that the fixed-effects specification captures most of the variation in firm-level market equity.

3.3 Table 2: Model B Predictive Benchmark (Random Forest vs. OLS)

Model	Train Period	Test Period	Test R^2	Test RMSE
OLS Benchmark	2002–2018	2019–2023	–0.329	0.0338
Random Forest	2002–2018	2019–2023	–0.192	0.0310
Naive Mean Baseline	2002–2018	2019–2023	–0.001	0.0271

All three models produce negative out-of-sample R^2 values, confirming that annual REIT returns are difficult to predict from the available features—a finding common in asset-return panels. The Random

Forest marginally outperforms OLS, but neither beats the naive mean. This benchmark confirms that the fixed-effects model is best interpreted as capturing long-run associations, not as a forecasting engine.

3.4 Diagnostics

Heteroskedasticity (Breusch-Pagan): LM statistic = 253.44 (p < 10[redacted]). Strong evidence of heteroskedasticity is present, which is why two-way clustered standard errors are used throughout rather than conventional OLS standard errors.

Multicollinearity (VIF): Maximum VIF = 11.84 (debt/assets); MORT_LAG2 VIF = 8.05; year trend VIF = 8.35. These elevated values are expected in an annual macro panel where the national rate series co-moves with broad time trends. Coefficients are interpreted with appropriate caution; the mortgage-rate and time-trend effects should not be interpreted in complete isolation.

Residual plots: The residuals-vs-fitted and Q-Q plots show non-ideal tails consistent with the Breusch-Pagan result. No systematic nonlinear pattern is visible in the residuals-vs-fitted plot, providing basic confirmation of model specification. All inference relies on the clustered standard errors rather than distributional assumptions.

3.5 Robustness Checks

Check	Coefficient	p-value	N	Takeaway
Lag 1 (baseline spec)	0.0024	0.552	2,540	Not significant
Lag 2 (main model)	–0.0007	0.896	2,428	Not significant
Lag 3 (alternate lag)	0.0085	0.138	2,327	Not significant
Exclude crisis years	–0.0000	0.999	1,992	No crisis-year artifact
Large REITs only	0.0003	0.959	1,215	No size concentration
Small REITs only	–0.0029	0.583	1,213	No size concentration

Across all lag specifications, crisis exclusion, and size subsamples, the mortgage-rate coefficient on annual returns remains statistically insignificant. This consistency strengthens confidence that the null result in the returns model is genuine rather than an artifact of any single modeling choice.

4. Conclusions & Recommendations

4.1 Investment Implications

Our findings support a rate-sensitive valuation framework for U.S. REITs. Because the mortgage rate effect is statistically significant and economically large in the market-equity model but absent in the returns model, the data suggest that rate shocks compress prices without proportionately reducing near-term income streams. This creates two distinct risk exposures for REIT investors:

- **Valuation risk (significant):** REITs with high leverage (debt/assets) and high book-to-market ratios face amplified market-cap compression during rate-hiking cycles. Overweight firms with conservative balance sheets.
- **Return-volatility risk (not rate-driven):** Near-term total returns are driven more by beta (systematic risk) than by rate levels. High-beta REITs should be sized accordingly based on portfolio risk

tolerance.

• **Sector tilt:** Underweight highly leveraged Office and Retail REITs where debt/assets are elevated. Overweight Industrial and Net-Lease REITs with lower leverage profiles and inflation-linked rent escalators that partially offset rate pressure.

• **Rate environment monitoring:** The two-year lag structure implies that the 2022–2023 hiking cycle's full valuation impact may not have been fully reflected in market prices through 2024. Monitor for continued multiple compression if rates remain elevated.

4.2 Risk Assessment

The primary model risk is **omitted variable bias.** The national mortgage-rate series captures broad rate conditions but does not separately identify credit-spread shocks, cap-rate changes driven by real estate supply dynamics, or REIT-sector-specific demand factors (e.g., e-commerce tailwinds for industrials, remote-work headwinds for offices). If these factors correlate with the mortgage rate across the sample period, our coefficient estimates may attribute their effects to the rate channel. The insignificant return result in particular may reflect omitted sentiment or liquidity-premium factors.

A secondary risk is **non-stationarity.** Both the mortgage-rate series and log market equity exhibit trend-like behavior over 2000–2023. The year trend and entity fixed effects partially address this, but the elevated VIFs on MORT_LAG2 and TREND suggest residual collinearity. Investors should not extrapolate the coefficient magnitudes mechanically to future rate scenarios.

4.3 Caveats and Limitations

• **Annual aggregation:** Annual-average returns and annual-average mortgage rates smooth high-frequency dynamics. Monthly or quarterly panels might reveal stronger or different short-run rate effects that are obscured here.

• **Parallel trends assumption:** The entity FE model assumes that, in the absence of rate changes, each REIT's valuation would follow a common trend (captured by TREND + entity FE). If certain REIT types (e.g., Office vs. Industrial) were on diverging structural paths, this assumption may be violated.

• **External validity:** Results are specific to U.S. REITs over 2000–2023, a period that includes two major crises and an unprecedented rate-hiking cycle. Caution is warranted when applying these estimates to other real estate markets or time periods.

• **Strict TWFE identification:** The national mortgage-rate variable is absorbed under full year fixed effects, requiring the year-trend approach. While this is econometrically sound, it means our identification relies on variation around a linear time trend rather than the full richness of year-by-year variation.

5. References

Federal Reserve Economic Data (FRED). (2024). *15-Year Fixed Rate Mortgage Average in the United States (MORTGAGE15US).* Federal Reserve Bank of St. Louis. https://fred.stlouisfed.org/series/MORTGAGE15US

REIT Master Dataset. (2024). *REIT Sample 2000–2024: All Variables.* data/raw/REIT_sample_2000_2024_All_Variables.csv [Course-provided dataset, QM 2023 Capstone, University of Tulsa].

Seagraves, C. (2026). *QM 2023: Statistics II — Capstone Project Guidelines.* University of Tulsa, Spring 2026.

Breusch, T. S., & Pagan, A. R. (1979). A simple test for heteroscedasticity and random coefficient variation. *Econometrica, 47*(5), 1287–1294.

Hausman, J. A. (1978). Specification tests in econometrics. *Econometrica, 46*(6), 1251–1271.

Appendix: AI Use Audit

In accordance with course policy, this appendix documents AI tool use across all four milestones of the capstone project. All AI-assisted outputs were reviewed, verified against source data, and critically evaluated by team members before inclusion.

Milestone 1 — Data Pipeline

• **Tool:** Claude (Anthropic). Used to debug pandas merge logic and identify duplicate-key issues in the REIT–FRED panel join.

• **Prompt example:** 'Our merge is producing duplicate rows — here is the head of each dataframe. What merge key or type issue would cause this?' Output was reviewed and the identified key mismatch (string year vs. integer year) was verified manually.

• **Verification:** Final row count and key uniqueness confirmed via value_counts() and assert statements in merge_final_panel.py.

Milestone 2 — Exploratory Analysis

• **Tool:** Claude. Used to suggest visualization approaches for the dual-axis rate-vs-return time series and to draft initial lag-correlation interpretation language.

• **Critique applied:** AI suggestion to emphasize lag-2 correlation ($r = -0.09$) was tempered after team review found the correlation modest and not robust across subperiods.

Milestone 3 — Econometric Models

• **Tool:** Claude. Consulted on the strict TWFE absorption issue — specifically whether including a national annual regressor alongside full year dummies is identified. AI explanation confirmed the collinearity logic; the team verified this independently by running the strict TWFE spec and observing the dropped variable.

• **Tool:** Claude. Used to help format regression output tables from raw Python dictionaries into the markdown table structure submitted in M3_econometric_models.md.

• **Verification:** All numerical values in the memo tables were cross-checked against results/tables/M3_regression_table.csv generated directly by capstone_models.py.

Milestone 4 — Investment Memo

• **Tool:** Claude. Used to assist with drafting and structuring the M4 investment memo based on the team's M3 numerical outputs and M2 visualizations.

• **Team accountability:** All economic interpretations, investment recommendations, and limitations language were reviewed and approved by team members. No numerical values were AI-generated; all figures are drawn directly from the capstone code outputs.

• **Critique applied:** Initial AI-drafted recommendation to 'underweight all REITs' was revised by the team to a more nuanced sector-tilt recommendation grounded in the leverage and beta coefficients from the model.

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